

# Chenfeng Huang

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## RESEARCH INTERESTS

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Interpretable machine learning, time series analysis, PAC-Bayesian learning, out-of-distribution generalization, and causal machine learning.

## EDUCATION

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### University of California, Los Angeles

*Ph.D. in Statistics*

- Advisor: George Michailidis

**2026 - Present**

*Los Angeles, CA*

### University of Southern California

*M.S. in Applied Data Science*

- GPA: 4.0/4.0

**2023 - 2025**

*Los Angeles, CA*

### Santa Clara University

*B.S. in Computer Science*

- GPA: 3.5/4.0

**2020 - 2022**

*Santa Clara, CA*

## RESEARCH EXPERIENCE

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### Research Assistant

*University of California, Los Angeles*

- Developing a model-agnostic online PAC-Bayesian calibration framework for time series forecasting under distribution shift, combining martingale certificates with gated residual Bayesian correction to improve stability under covariate and concept shift.
- Investigating few-shot identification of linear dynamical systems through PAC-Bayesian meta-learning, deriving generalization guarantees under sequential dependence and enabling Bayesian adaptation from short, noisy trajectories.

**2026 - Present**

*Los Angeles, CA*

### Research Assistant

*North Carolina State University*

- Proposed an expert-guided causal Bayesian optimization framework with adaptive expert weighting, confidence-based handover, and safety checks to improve intervention quality in complex causal environments.
- Developed a causal jump-diffusion method for time-series anomaly detection that models latent environment shifts and contrasts factual and counterfactual trajectories for robust detection.
- Contributed to ongoing work on causal Bayesian optimization foundations and PAC-Bayesian methods for time-series generalization under dependence and distribution shift.

**2024 - Present**

*Raleigh, NC*

## SELECTED PUBLICATIONS

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- **C. Huang**, Z. Ma, and G. Michailidis. "Model-Agnostic Online Certificate-Driven Calibration for Time Series Forecasting Under Distribution Shift." Under review, 2026.
- **C. Huang** and G. Michailidis. "PAC-Bayesian Meta-Learning for Few-Shot Identification of Linear Dynamical Systems." Under review, 2026.
- Z. Ma, **C. Huang**, T. T. Le, and H. Tran. "Confidence-Based Handover for Causal Bayesian Optimization with Adaptive Expert Trust." Under review, 2026.
- Z. Ma and **C. Huang**. "A Causal Perspective on Jump-Diffusion for Time-Series Anomaly Detection." Under review, 2026.
- Z. Ma and **C. Huang**. "Effective Series Decomposition and Components Learning for Time Series Generation." *IEEE International Conference on Data Mining (ICDM)*, 2025.

## PRESENTATIONS

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- **Joint Mathematics Meetings (JMM)**, 2026. "Expert-guided Causal Bayesian Optimization." Abstract accepted for presentation.
- **IEEE International Conference on Data Mining (ICDM)**, 2025. "Effective Series Decomposition and Components Learning for Time Series Generation."

## TEACHING EXPERIENCE

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## Mathematics Tutor

Sierra College

2018

Rocklin, CA

- Tutored Calculus I and Calculus II students in one-on-one and small-group settings.
- Completed the College Reading and Learning Association (CRLA) Level I tutor training program.

## SELECTED HONORS

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- Kaggle Silver Medal, *Detect Behavior with Sensor Data*, rank 16/2657 (top 1%), 2025.
- Kaggle Silver Medal, *Charting Student Math Misunderstandings*, rank 92/1858 (top 5%), 2025.
- Kaggle Silver Medal, *Child Mind Institute Problematic Internet Use*, rank 111/3599 (top 3%), 2024.

## TECHNICAL SKILLS

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- **Languages:** Python, C/C++, Java, SQL
- **Machine Learning and Data:** PyTorch, TensorFlow, MLflow, Hadoop, MongoDB
- **Tools:** Linux, Git